

Angles in the Coordinate Plane

Standard Position

DEFINITION: Standard Position, Quadrantal and Coterminal Angles

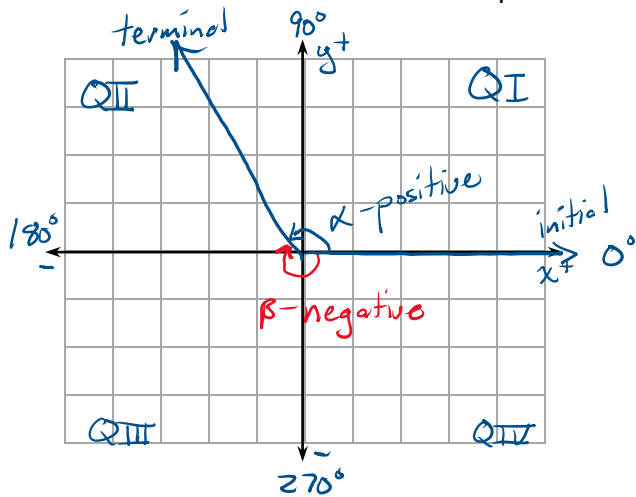
An angle is said to be in standard position if the vertex is positioned at the origin with the initial side along the positive x -axis.

Angles formed by clockwise rotation are negative in measure, while angles formed by counterclockwise rotation are positive in measure.

Any two angles in standard position, that share a common terminal side are said to be coterminal angles. Coterminal angles differ by a multiple of 360° .

Any angle, in standard position, with terminal side along either the x - or y -axis is called a quadrantal angle.

Use the coordinate axis below to provide a demonstration of the terminology defined above.



$$\alpha \in QII$$

terminal side for α falls in QII

$\alpha > 0$, counterclockwise rotation

$\beta < 0$, clockwise rotation

α and β differ by 360°

α and β are coterminal angles!

EXAMPLE 1

Find two coterminal angles, one negative and one positive, for each angle θ .

(a) $\theta = 260^\circ$

$$\begin{array}{r} 260 \\ -360 \\ \hline -100^\circ \end{array} \quad \begin{array}{r} 260 \\ +360 \\ \hline 620^\circ \end{array}$$

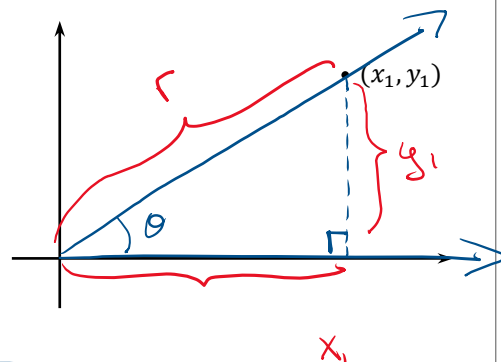
(b) $\theta = -944^\circ$

$$\begin{array}{r} -944 \\ +360 \\ \hline -584^\circ \\ +360 \\ \hline -224^\circ \\ +360 \\ \hline 136^\circ \end{array} \leftarrow \text{coterminal angles!}$$

EXAMPLE 2

Let θ be a positive acute angle, in standard position, whose terminal side passes through the point, (x_1, y_1) . Complete the following:

- Graph θ .
- Use the given point and the origin to construct a right triangle with base along the positive x -axis.
- Let r be the distance from the origin to the point (x_1, y_1) . Find this distance.
- Using right triangle trigonometry, evaluate $\sin(\theta)$, $\cos(\theta)$, and $\tan(\theta)$ in terms of x_1, y_1 , and r .



$$r^2 = (x_1)^2 + (y_1)^2 \rightarrow r = \sqrt{(x_1)^2 + (y_1)^2}$$

SOH $\sin(\theta) = \frac{\text{opp}}{\text{hyp}} = \frac{y_1}{r}$

CAH $\cos(\theta) = \frac{\text{adj}}{\text{hyp}} = \frac{x_1}{r}$

TOA $\tan(\theta) = \frac{\text{opp}}{\text{adj}} = \frac{y_1}{x_1}$

In the above example, we see that for an angle in quadrant I, with terminal side passing through some point, (x_1, y_1) , the sine, cosine, and tangent values for this angle can be defined as ratios of x_1, y_1 , and r , where $r = \sqrt{(x_1)^2 + (y_1)^2}$. However, when working with angles outside quadrant I (greater than 90°), our right triangle definitions for sine, cosine, and tangent become inadequate since we are no longer dealing with right triangles. Fortunately, the relationships we discovered in Example 2 hold true in any quadrant!

Defining Sine, Cosine, and Tangent for any Angle

DEFINITION: Sine, Cosine, and Tangent Values for any Angle

Let θ be any angle in standard position with terminal side passing through the point, $P(x, y)$, and distance from the origin given by r , where $r = \sqrt{x^2 + y^2}$, $r > 0$. Then, the sine, cosine, and tangent values for θ are defined as follows:

$$\sin(\theta) = \frac{y}{r} \qquad \cos(\theta) = \frac{x}{r} \qquad \tan(\theta) = \frac{y}{x}, x \neq 0$$

EXAMPLE 3

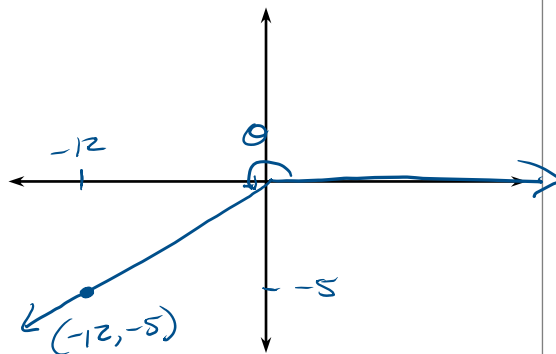
Draw an angle θ , in standard position, with terminal side passing through the point $(-12, -5)$. Evaluate the sine, cosine, and tangent values for θ .

$$\theta \in \text{Q III}$$

$$\sin(\theta) = \frac{y}{r} = \frac{-5}{13} \quad \text{negative}$$

$$\cos(\theta) = \frac{x}{r} = \frac{-12}{13} \quad \text{negative}$$

$$\tan(\theta) = \frac{y}{x} = \frac{-5}{-12} = \frac{5}{12} \quad \text{positive}$$



$$\begin{aligned} x &= -12, y = -5 \\ r &= \sqrt{(-12)^2 + (-5)^2} \\ &= \sqrt{169} \\ &= 13 \end{aligned}$$

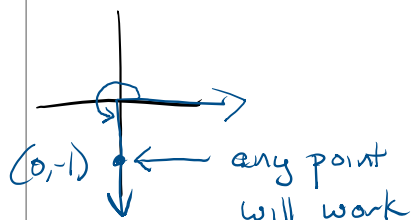
EXAMPLE 4

Evaluate the sine, cosine, and tangent values for each angle θ in standard position.

(a) Terminal side of θ passing through $(-\pi, 0)$ $\theta = 180^\circ \rightarrow x = -\pi, y = 0, r = \pi$

$$\sin \theta = \frac{0}{\pi} = 0, \quad \cos \theta = \frac{-\pi}{\pi} = -1, \quad \tan \theta = \frac{0}{\pi} = 0$$

(b) $\theta = 270^\circ$



$$\sin \theta = \frac{-1}{1} = -1$$

$$\cos \theta = \frac{0}{1} = 0$$

$$\tan \theta = \frac{-1}{0} = \text{undefined!}$$

EXAMPLE 5

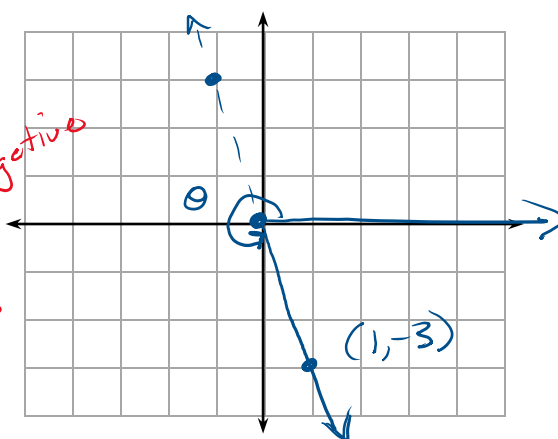
Let θ be the least positive angle in standard position, with terminal side defined by $y = -3x$, with $x > 0$. Graph this angle and evaluate $\sin(\theta)$, $\cos(\theta)$, and $\tan(\theta)$.

$$y = -3x + 0, m = -\frac{3}{1}, b = 0$$

$$\sin \theta = \frac{y}{r} = \frac{-3}{\sqrt{10}} = \frac{-3\sqrt{10}}{10}$$

$$\cos \theta = \frac{x}{r} = \frac{1}{\sqrt{10}} = \frac{\sqrt{10}}{10}$$

$$\tan \theta = \frac{y}{x} = \frac{-3}{1} = -3$$



$$x = 1, y = -3$$

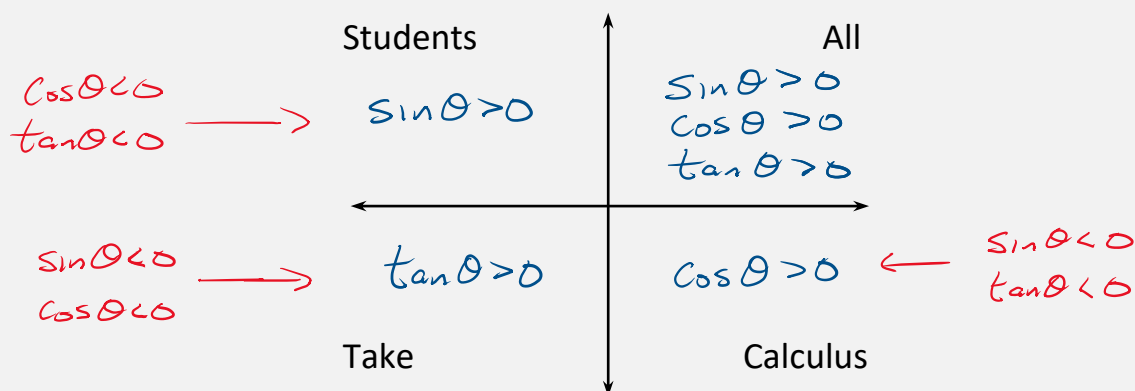
$$r = \sqrt{(1)^2 + (-3)^2}$$

$$r = \sqrt{10}$$

Now that we are evaluating sine, cosine, and tangent for any angle, not just acute angles of a right triangle, you may have noticed our results are sometimes positive and other times negative. This is an important observation, and we use the mnemonic, All Students

Take Calculus to help us recall the sign for any trigonometric function by knowing the quadrant for the angle.

All Students Take Calculus



EXAMPLE 6

Determine the sign for each expression.

(a) $\sin(190^\circ) < 0$

$190^\circ \in Q_{III}$

(b) $\cos(337^\circ) > 0$

$337^\circ \in Q_{IV}$

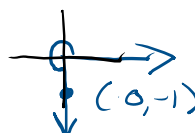
S	A
T	C

(c) $\tan(122^\circ) < 0$

$122^\circ \in Q_{II}$

(d) $\tan(270^\circ) = \text{undefined}$

270° is quadrantal angle

**EXAMPLE 7**

Identify all possible quadrants for the angle θ , in standard position, satisfying the given conditions.

(a) $\sin(\theta) > 0, \cos(\theta) < 0$

$\sin \theta > 0 \rightarrow QI, QII$

$\cos \theta < 0 \rightarrow QII, QIII$

$\theta \in QII$

(b) $\tan(\theta) > 0, \cos(\theta) > 0, \sin(\theta) < 0$

$\tan \theta > 0 \rightarrow QI, QIII$

$\cos \theta > 0 \rightarrow QI, QIV$

$\sin \theta < 0 \rightarrow QIII, QIV$

no such θ exists!

EXAMPLE 8

So far, we've defined three of the six trigonometric ratios, sine, cosine, and tangent. The others, cosecant, secant, and cotangent, are just reciprocals of the three we know.

$$\begin{array}{ll} \text{sine: } \sin(\theta) = \frac{y}{r} & \rightarrow \quad \text{cosecant: } \csc(\theta) = \frac{r}{y} \\ \text{cosine: } \cos(\theta) = \frac{x}{r} & \rightarrow \quad \text{secant: } \sec(\theta) = \frac{r}{x} \\ \text{tangent: } \tan(\theta) = \frac{y}{x} & \rightarrow \quad \text{cotangent: } \cot(\theta) = \frac{x}{y} \end{array}$$

- (a) Given θ is an angle in standard position, with terminal side passing through the point $(-12, -5)$. Evaluate the secant, cosecant, and cotangent values for θ . (Hint: See Example 3.)

$$\sin \theta = \frac{-5}{13} \rightarrow \csc \theta = \frac{-13}{5}$$

$$\cos \theta = \frac{-12}{13} \rightarrow \sec \theta = \frac{-13}{12}$$

$$\tan \theta = \frac{-5}{-12} \rightarrow \cot \theta = \frac{-12}{-5} = \frac{12}{5}$$

$$\begin{array}{l} x = -12 \\ y = -5 \\ r = 13 \text{ (ex 3)} \end{array}$$

- (b) Determine the sign of the expression, $\sin(25^\circ) \cdot \cot(100^\circ) \cdot \sec(225^\circ)$.

$$\begin{array}{l} \sin(25^\circ) \cdot \cot(100^\circ) \cdot \sec(225^\circ) \\ (+) \cdot (-) \cdot (-) = \text{positive result!} \\ \begin{array}{l} \nearrow 25^\circ \in \text{Q I} \quad \nearrow 100^\circ \in \text{Q II} \quad \nearrow 225^\circ \in \text{Q III} \\ \nearrow \tan(100^\circ) < 0 \quad \nearrow \cos(225^\circ) < 0 \end{array} \end{array}$$

- (c) Identify all possible quadrants for the angle θ , such that $\sin(\theta) > 0$ and $\cot(\theta) < 0$.

$$\sin \theta > 0 \rightarrow \theta \in \text{Q I}, \theta \in \text{Q II}$$

$$\cot \theta < 0 \rightarrow \theta \in \text{Q II}, \theta \in \text{Q IV}$$

$$\nearrow \tan \theta < 0$$

$$\theta \in \text{Q II}$$

